

No. of Functional Features Included in the Solution

Date	4 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

The Credit Card Approval Prediction Framework integrates a series of core functional modules engineered to automate screening workflows and minimize manual processing friction. By mapping out clear operational boundaries—ranging from automated raw data cleansing and categorical feature encoding to live classification scoring and multi-model performance validation—this overview details how each structural component transforms unstructured applicant metrics into immediate, data-driven underwriting decisions.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Interactive Application Form	Allows users or bank staff to input applicant details like age, income, and employment status.	Frontend UI Component	Done	15%
2	Automated Missing Value Handling	Automatically cleanses incoming datasets by imputing missing values or structural anomalies.	Backend Data Preprocessor	Done	10%
3	Categorical Feature Encoder	Transforms non-numeric variables (e.g., occupation type, education) into mathematical arrays.	Core Data Pipeline	Done	10%
4	Risk Assessment Engine	Evaluates applicant data tables through a trained classifier model to establish risk scores.	Machine Learning Module	Done	25%
5	Live Status Indicator	Spits out immediate binary classification metrics (Approved or Rejected) directly to the active screen interface.	API Response Handler	Done	15%
6	Threshold Adjustment Controller	Provides a safety cushion matrix to adjust internal risk confidence boundaries during economic changes.	Administration Panel	In Progress	10%
7	Secure Log Tracking Storage	Records historical requests, scoring values, and operational time markers safely to local file targets.	Administration Panel	Done	10%
8	Multi-Model Evaluation Dashboard	Compares different algorithm predictions (XGBoost vs Random Forest) to confirm structural accuracy before runtime integration.	Diagnostic Engine	Pending	5%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Input Collection Panels, Decision Portals	HTML Web Applications, Error-Checking Validation Pop-Ups
2	Backend / Logic	Preprocessing Pipelines, Classification Scoring	Data Tokenizers, Serialized Machine Learning Model Evaluators
3	Database / Storage	Local Dataset Repositories, Diagnostic Files	Tabular Data CSV Stores, Performance Optimization System Logs
4	API / Integration	Communication Routes, Pipeline Controls	Flask Endpoint Wrappers, JSON Request / Response Modules
5	Security / Authentication	Access Auditing, Form Field Interceptors	Internal Type Verification Frameworks, Malicious Character Input Protectors